

Local Protocol #: 4T-03-1

Protocol Title: A PHASE II STUDY OF GEMCITABINE, OXALIPLATIN,
AND PACLITAXEL IN PATIENTS WITH REFRACTORY GERM CELL
CARCINOMA

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CLINICAL PROTOCOL NUMBER: 4T-03-1

TITLE: A PHASE II STUDY OF GEMCITABINE,
OXALIPLATIN, AND PACLITAXEL IN PATIENTS
WITH REFRACTORY GERM CELL CARCINOMA

VERSION DATE: 07/20/2007

SITE: TESTIS

HISTOLOGY: GERM CELL TUMOR

STAGE: PROGRESSIVE, METASTATIC, REFRACTORY

MODALITY: CHEMOTHERAPY

TYPE: PHASE II

ARMS: NON-RANDOMIZED

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AMENDMENTS/REVISIONS:

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OF THE EXPERIMENTAL PROTOCOL OUTSIDE THE PARTICIPATING INSTITUTIONS

1.0 OBJECTIVES

- ___ To assess the response rate (confirmed complete and partial responses) in patients with germ cell malignancies who have failed initial cisplatin containing chemotherapy utilizing the combination of oxaliplatin, gemcitabine, and paclitaxel.
- ___ To assess the overall survival and progression free survival in this group of patients.
- ___ To evaluate the qualitative and quantitative toxicities of this regimen.

2.0 BACKGROUND AND HYPOTHESES

The expectation for cure with germ cell cancer exceeds ninety percent, and 70-90% of patients are cured with a combination of primary chemotherapy with or without surgery. The prognosis is much worse for those patients progressing or relapsing post primary chemotherapy. Poor risk disease is especially disappointing, given its 50% incidence of failure of durable complete response.¹

2.1 No satisfactory therapy is available for patients with intensively pretreated germ cell neoplasms.

Refractory germ cell cancer provides an opportune setting for the evaluation of novel chemotherapeutic combinations. Although high cure rates can be achieved with first line therapy, there is currently a need for more effective alternatives in the setting of refractory and metastatic disease. Progress during or relapse after therapy portends an extremely poor prognosis. The onset of cisplatin resistance, a staple of the majority of germ cell regimens, further complicates treatment. No truly effective treatment is available for patients who are refractory to front-line platinum-based chemotherapy. Intensively treated patients face a similar situation. Additionally, the subsets of patients who represent late relapse (>2 years) are included in any evaluation of poor prognosis disease.

Current applied regimens for salvage therapy can offer the patient long-term survival in a minority of cases with only an expected twenty-five percent chance of complete response with second line standard therapy and 15% with third line therapy using high dose chemotherapy and stem cell transplant.² There exists a need for unique regimens offering a solution to the previously noted conundrums. Recently, responses have been noted in germ cell tumors with use of new chemotherapeutics, paclitaxel, gemcitabine, and oxaliplatin.³⁻⁹ The current protocol proposes a formal phase II trial to evaluate the role of gemcitabine, oxaliplatin, and paclitaxel in relapsed and/or cisplatin-refractory germ cell cancer.³

2.2 Gemcitabine has single agent activity in germ cell cancer

The anti-tumor activity and toxicity of gemcitabine have been studied only preliminarily in the setting of germ cell cancer. Of note, there have been responses observed in patients with previous high dose therapy and/or cisplatin refractoriness. Studies done at Indiana University and by the German Testicular study group have demonstrated a 15-19% response rate with minimal grade 3 & 4 toxicities. Overall, the non-hematological toxicities were mild with fever, nausea, and vomiting comprising the main side effects. In patients with high dose chemotherapy and stem cell support the hematologic toxicity was greater. In a study of patients with relapsed or cisplatin refractory germ cell tumors. Four (13%) patients experienced grades 3& 4 granulocytopenia with 7 (22%) experiencing grades 3& 4 thrombocytopenia. Phase I studies with combinations of gemcitabine and paclitaxel showed acceptable toxicity. In phase II studies, the combination proved applicable with 21% responders and 4 pathologically documented complete responders.^{-4 5}

Activity of single agent gemcitabine in relapsed/refractory germ cell tumors

Study	Gemcitabine mg/m2	No	prev HDCT	DDP refractory	RR	Median Survival
Bokemeyer (1999)	1,000 d1,8,15 q28	31	22 (71%)	17 (54%)	6(19%)	6mo (2-23)
Einhorn (1999)	1,000 d1,8,15 q28	20	11 (55%)	13 (65%)	3(15%)	Not given
Total		51	33 (65%)	30 (58%)	9(17%)	

2.3 Oxaliplatin has evidenced role in germ cell cancer⁶

Oxaliplatin offers a solution to a unique dilemma in the treatment of refractory disease: the existence of cisplatin refractoriness and neurotoxicity. It lacks the degree of ototoxicity, nephrotoxicity, myelotoxicity, and alopecia associated with cisplatin. Additionally, the diaminocyclohexane moiety on the compound relates a lack of cross-resistance to cisplatin perhaps by circumventing the glutathione resistance pathway. In vitro studies have indicated incomplete cross-resistance to cisplatin – the standard of all regimens. Although its dose limiting toxicity of neurotoxicity is similar, it offers a spectrum of activity different from other platinum agents. In-vitro studies have proven that oxaliplatin cytotoxicity is significantly superior to cisplatin in cell lines with both acquired and intrinsic intermediate level resistance to cisplatin. Both paclitaxel and gemcitabine have shown significant activity in refractory male germ cell tumors and both have exhibited synergism with cisplatin. In phase II studies, oxaliplatin had an overall 13% response rate with 4 patients exhibiting partial response. Responses were seen in patients who were late relapses and those who had failed high dose therapy. Supra-additive effects have been noted with the combination of gemcitabine and oxaliplatin in human leukemia and colorectal cancer cell lines, indicating a potential advantage in our population.^{7 8}

Activity of single agent Oxaliplatin in relapsed/refractory germ cell tumors

Study	Oxaliplatin mg/m2	No	prev HDCT	DDP refractory	Late Relapse	RR
Kollmansberger (2002)	60 d1,8,15 q28 or 130 d1,15 q28	32	78%	85%	34%	13%
(percentage of patients in study)						

2.4 Paclitaxel has evidenced role in germ cell cancer⁹

Activity of single agent Paclitaxel in relapsed/refractory germ cell tumors

Study	Paclitaxel mg/m2	No	Responses
Bokemeyer (1994)	135-225	10	3 PR (30%)
Motzer (1994)	250 + GCSF	31	3 CR 5PR (26%)
Nasario (1995)	250 + GCSF	15	0 CR 2 PR (13%)
Bokemeyer (1996)	225	3	2 PR (11%)
Total		98	5 CR 16 PR (21%)

3.0 DRUG INFORMATION

3.1 Oxaliplatin (ELOXATIN)¹⁰

Oxaliplatin (trans-*l*-1,2-diaminocyclohexane oxalatoplatinum), is a novel antineoplastic platinum derivative with a 1,2-diaminocyclohexane [DACH] carrier ligand. Although the precise mechanism of action is unknown, platinum compounds in general are thought to exert their cytotoxic effects through the formation of DNA adducts that block both DNA replication and transcription, resulting in cell death in actively dividing cells as well as the induction of apoptosis. Oxaliplatin is more potent than cisplatin *in vitro* requiring fewer DNA adducts to achieve an equal level of cytotoxicity¹¹ and showed efficacy in preclinical studies against many tumor cell lines^(12,13,14,15,16,17,18) including some that are resistant to cisplatin and carboplatin.

3.1.1 Mechanism of Action

Like cisplatin, oxaliplatin reacts with DNA, forming mainly platinated intrastrand links with two adjacent guanines or a guanine adjacent to an adenine¹⁹. However, DACH-platinum adducts formed by oxaliplatin are apparently more effective at inhibiting DNA synthesis⁽²⁰⁾ and are more cytotoxic than cis-diammine-platinum adducts formed from cisplatin and carboplatin²¹. The 1,2-DACH carrier ligand effects the rate of mono- and diadduct conversion²². It also effects the ability of cells to tolerate unrepaired platinum-DNA adducts²³. The latter effect may be responsible for differentiating the antineoplastic activity of oxaliplatin from those of carboplatin or cisplatin. The two mechanisms that reproducibly discriminate between cisplatin and oxaliplatin are defects in mismatch repair and enhanced replicative bypass (13). Several studies have demonstrated that the loss of the mismatch repair (MMR) enzyme complex 1 is a contributor to intrinsic resistance to cisplatin, but not oxaliplatin.

3.1.2 Preclinical Studies

In vitro, oxaliplatin has shown antiproliferative activity equivalent to or higher than that of cisplatin against murine and human cancer cell lines. These cell lines include non-seminomatous germ cell cancer cell lines. In nonseminomatous germ cell cancer cell lines with either acquired (H12DDP clone) or intrinsic (1777Nrp Cl-A clone) resistance to cisplatin, oxaliplatin was significantly more cytotoxic than cisplatin. These data suggest that the mechanism of action of oxaliplatin differs from that of cisplatin or carboplatin. Not all cisplatin-resistant cell lines are sensitive to oxaliplatin.

In vitro and *in vivo*, oxaliplatin has shown additive or synergistic cytotoxic properties with fluoropyrimidines (5-FU and gemcitabine), thymidylate synthase inhibitors (AG337), topoisomerase I inhibitors (CPT-11 and SN-38), microtubule inhibitors (paclitaxel), and DNA modifying/alkylating agents (cisplatin and cyclophosphamide). Although oxaliplatin demonstrates additive/synergistic activity in combination with many standard anticancer agents in preclinical models, some combinations do not demonstrate increased *in vivo* activity and are sometimes associated with increased toxicity.

3.1.3 Clinical Pharmacokinetics and Safety Profile²⁴

The safety profile of oxaliplatin was evaluated in a phase I intra-patient escalation study of 44 patients with advanced cancer, who received 116 courses of oxaliplatin through seven levels, from 45 mg/m² to 200 mg/m² every four weeks. Initially, the duration of administration was 1 hour and was increased to 6 hours for doses of >60 mg/m² in an effort to reduce gastrointestinal toxicity. Most patients had received prior chemotherapy, including cisplatin. Oxaliplatin was administered without prehydration or post hydration. All patients experienced nausea and vomiting, as reported in an earlier study by Mathé et al. As a result, systematic pretreatment with antiemetics was given to all patients receiving 390 mg/m² oxaliplatin, which reduced Grade 3 or 4 nausea and vomiting to 11%. Diarrhea was mild (primarily Grade 1 or 2 in 24% of therapy courses).

Hematologic toxicity was moderate. Grade 1 or 2 thrombocytopenia was dose related and occurred in 13% of patients receiving 135 to 150 mg/m², and 28.5% of patients receiving 175 to 200 mg/m². Similarly, only Grade 1 or 2 neutropenia was observed, and hemoglobin levels remained unchanged. The dose-limiting side-effect of oxaliplatin therapy was a transient peripheral neuropathy manifesting as paresthesia in the fingers, hands, toes and lips, as well as dysesthesias in the forearms, legs, mouth and throat triggered or enhanced by exposure to cold. This toxicity usually appeared at doses 390 mg/m² and affected up to 75% of patients treated with 200 mg/m². The duration and intensity of the symptoms increased with the number of courses administered. Thus, neurologic toxicity observed in this trial was cumulative. Grade 3 neurologic toxicity occurred in 6% of cycles and 14% of patients and slowly regressed with symptoms disappearing after 6 months. The recommended phase II dose was 135 mg/m² administered over at least 1 hour every four weeks.

3.1.4 Rationale

It has been noted in the recent years that platinum complexes with the DACH carrier ligand were much more effective against a variety of cisplatin-sensitive and cisplatin-resistant cell lines of both murine and human origin. Oxaliplatin is a new platinum analog of the DACH family. In human cell lines, oxaliplatin is active against colon, ovarian, breast, melanoma, bladder, glioma, erythroleukemia, lung, neuroblastoma and nonseminomatous testicular cells. Oxaliplatin had greater activity than cisplatin in approximately 40% of the human cell lines tested. Oxaliplatin has shown equal or superior *in vivo* antitumor activity when compared to cisplatin, in a variety of tumor models, including cisplatin-resistant tumors L1210/DDP, LGC and MA-16c. When combined with other cytotoxic agents (5FU, SN38, CDDP, carboplatin, taxanes), oxaliplatin has additive and/or synergistic antitumoral effects on various *in vitro* and *in-vivo* models. Oxaliplatin is found to be 5 times less toxic than cisplatin. A cumulative, dose-related and reversible peripheral sensory neuropathy was noted as the dose-limiting toxicity. Oxaliplatin induced a peripheral sensory neuropathy characterized by dysesthesia and/or distal paresthesia, induced or exacerbated by cold and generally regressing between cycles of treatment. The symptoms generally appeared following treatment at doses greater than 100 mg/m²/cycle. When treatment is continued, the extent and duration of symptoms may increase, becoming permanent between cycles. Other toxicities include gastrointestinal toxicity characterized by nausea, vomiting and diarrhea, minimal hematotoxicity, nephrotoxicity and of clinical auditory toxicity. Ototoxicity occurs in less than 1% of patients and nephrotoxicity is rare. Myelosuppression is usually mild.

3.2 Gemcitabine HCL (GEMZAR)

3.2.1 Description:

Gemcitabine (difluorodeoxycytidine), an analog of cytosine arabinoside (Ara-C) is a novel pyrimidine anti-metabolite. The mechanism of action of gemcitabine has been well characterized. It can be deaminated to difluorodeoxyuridine by deoxycytidine deaminase or converted to dFdCMP by deoxycytidine kinase. Difluorodeoxyuridine is inactive, while dFdCMP is further metabolized to dFdCDP and dFdCTP, which, when incorporated into DNA, results in chain termination. In comparison to Ara-C incorporation into DNA, dFdCTP is less readily excised from DNA by DNA exonuclease. Thus, dFdCTP accumulates intracellularly to a greater degree than Ara-C. This may account, in part, for its different spectrum of pre-clinical and clinical activities. In addition, gemcitabine inhibits ribonucleotide reductase, an enzyme that produces deoxynucleotides that are required for DNA synthesis. Gemcitabine activity has been demonstrated in breast, ovarian, bladder, esophageal, and lung carcinoma.

3.2.2 Drug Administration, Storage, and Stability:

Gemcitabine is supplied as a lyophilized powder in sterile vials containing 200 mg or 1.0 gm of gemcitabine as the hydrochloride salt (expressed as the free base), mannitol, and sodium acetate. The lyophilized product should be stored below 30°C. To reconstitute, add 5 ml of 0.9% sodium chloride to the 200 mg vial or 25 ml of 0.9% sodium chloride to the 1.0 gm vial. These dilutions each yield a gemcitabine concentration of 38 mg/ml. An appropriate amount of drug will be prepared with normal saline and administered as a continuous infusion for 30 minutes. Once the drug has been reconstituted, it should be stored at room temperature and used within 24 hours.

3.2.3 Toxicity:

Initial phase I studies using a short infusion schedule of gemcitabine given weekly for 3 successive weeks followed by a 1 week rest established 790 mg/m²/week as the maximum tolerated dose (MTD) with dose-limiting toxicity consisting of myelosuppression (thrombocytopenia more so than granulocytopenia). More recent phase I/II studies have established 1250 mg/m² as a well tolerated dose. Principal toxicities reported were hematologic with World Health Organization (WHO) grade 4 neutropenia/ thrombocytopenia occurring rarely; reversible elevation in hepatic transaminases, proteinuria, mild skin rash with and without pruritis, and nausea/vomiting. A recent review of 201 patients treated with 1250 mg/m² on days 1, 8, and 15 q 28 days who had not received prior chemotherapy revealed WHO grade 3 and 4 neutropenia in 23% and 6%, respectively, WHO grade 3 proteinuria in less than 1%, WHO grade 3 nausea/ vomiting in 10%, and mild skin rash in 26%, with pruritus occurring in 10%.

3.3 Paclitaxel (TAXOL)

3.3.1. Description

Paclitaxel is a diterpene plant product found in the needles and bark of the western yew product with anti-tumor activity. Paclitaxel is obtained via an extractive process from *Taxus brevifolia* or *Taxus yunnanensis*. Its chemical name is (2aR,4S,4aS,6R,9S,11S,12S,12aR,12bS)-1,2a,3,4,4a,6,9,10,11,12,12a,12b-Dodecahydro-4,6,9,11,12,-12b-hexahydroxy-4a,8,13,13-tetramethyl-7,11-methano-5H-cyclodeca [3,4] benz [1,2-b] oxet-5-one 6,12b diacetate, 12-benzoate, 9 ester with (2R,3S) -N-benzoyl-3-phenylisoserine.

3.3.2 Mechanism of Action

Paclitaxel is a novel microtubule agent that promotes the assembly of microtubule agents from tubulin dimers and stabilizes microtubules by preventing depolymerization. This stability results in the inhibition of normal dynamic reorganization of the microtubule network that is essential for vital interphase and mitotic cellular functions. In addition paclitaxel induces abnormal arrays or “bundles” of microtubules throughout the cell cycle and multiple asters of microtubules during mitosis.

3.3.3 Toxicity :

Dose-limiting toxicity is myelosuppression with reversible granulocytopenia, anemia, and thrombocytopenia. Allergic reactions occur in up to 8% of patients receiving paclitaxel as an intravenous infusion over 6 to 24 hours. These can be acute anaphylactoid reactions, which include flushing,

hypotension, and bronchospasm; dermatitis and pruritus can be also observed. Hypertension has also been seen, and may be related to concomitant medication with dexamethasone. Premedication with diphenhydramine, cimetidine, and dexamethasone appears to diminish the incidence of these reactions.

a) Neural - Neurotoxicity can include distal painful paresthesias. Rarely, this toxicity has required discontinuation of drug due to pain, impairment of fine motor skills, or difficulty ambulating. Experience to date suggests that this neuropathy is reversible. Rarely, associated forms of neurotoxicity have included taste perversion, seizures, and mood changes. Some patients have reported vision abnormalities such as blurred vision, "flashing lights" and scintillating scotomata. Neuroencephalopathy has also been reported.

b) Gastrointestinal - Ischemic or infarcted colon, sometimes with involvement of other parts of the gastrointestinal tract, has also been seen. Patients reporting abdominal discomfort should be monitored closely. These events generally occurred while the patients were severely neutropenic. They may be most consistent with neutropenic enterocolitis (typhlitis). Although increased SGOT, SGPT, bilirubin and alkaline phosphatase, as well as hepatic failure and hepatic necrosis have been seen, one patient receiving this drug has also experienced hepatic encephalopathy, and two incidences of pancreatitis have been noted. Pulmonary toxicities that have occurred are pneumonitis and radiation pneumonitis (following concomitant paclitaxel and radiation).

c) Other non-hematologic reactions include: diarrhea, alopecia, myalgias and arthralgias, nausea or vomiting, mucositis (stomatitis and pharyngitis), lightheadedness, myopathy and fatigue. Less commonly, cardiotoxicity has been associated with paclitaxel administration, to include arrhythmias (sinus bradycardia, ventricular tachycardia, atrial arrhythmia, and heart block), and myocardial infarction. Skin reactions including erythema, induration, tenderness, ulceration, radiation recall, rash and nail changes have occurred including discoloration of fingernails and separation from nail bed.

Drug Accountability and Drug Ordering

Once the patient's eligibility has been established and the individual has been registered, the drugs will be supplied from commercial sources through the pharmacy of USC/Norris or LAC-USC Medical Center.

4.0 STAGING CRITERIA - Testis²⁵

PRIMARY TUMOR (T)

The extent of primary tumor is classified after radical orchiectomy.

- pTX Primary tumor cannot be assessed (if no radical orchiectomy has been performed, TX is used)
- pT0 No evidence of primary tumor (e.g., histologic scar in testis)
- pTis Intratubular germ cell neoplasia (carcinoma in situ)
- pT1 Tumor limited to the testis and epididymis without vascular/lymphatic invasion; tumor may invade into the tunica albuginea but not the tunica vaginalis
- pT2 Tumor limited to the testis and epididymis with vascular/lymphatic invasion, or tumor extending through the tunica albuginea with involvement of the tunica vaginalis
- pT3 Tumor invades the spermatic cord with or without vascular/lymphatic invasion
- pT4 Tumor invades the scrotum with or without vascular/lymphatic invasion

REGIONAL LYMPH NODES (N)

Clinical

- NX Regional lymph nodes cannot be assessed
- N0 No regional lymph node metastasis
- N1 Metastasis with a lymph node mass ≤ 2 cm in greatest dimension; or multiple lymph nodes, none more than 2 cm in greatest dimension
- N2 Metastasis with a lymph node mass, more than 2 cm but not more than 5 cm in greatest dimension or multiple lymph nodes, any one mass greater than 2 cm but not more than 5 cm in greatest dimension
- N3 Metastasis with a lymph node mass more than 5 cm in greatest dimension

Pathologic (pN)

- pNX Regional lymph nodes cannot be assessed

- pN0 No regional lymph node metastasis
- pN1 Metastasis with a lymph node mass, 2 cm or less in greatest dimension and less than or equal to 5 nodes positive, none more than 2 cm in greatest dimension
- pN2 Metastasis with a lymph node mass, more than 2 cm but not more than 5 cm in greatest dimension; or more than 5 nodes positive, none more than 5 cm; or evidence of extranodal extension of tumor
- pN3 Metastasis with a lymph node mass more than 5 cm in greatest dimension

DISTANT METASTASIS (M)

- MX Distant metastasis cannot be assessed
- M0 No distant metastasis
- M1 Distant metastasis
- M1a Nonregional nodal or pulmonary metastasis
- M1b Distant metastasis other than to nonregional lymph nodes and lungs

SERUM TUMOR MARKERS (S)

- SX Marker studies not available or not performed
- S0 Marker study levels within normal limits
- S1 LDH < 1.5 X N **AND**
hCG (mIU/ml) < 5,000 **AND**
AFP (ng/ml) < 1,000
- *S2 LDH 1.5-10 X N **OR**
hCG (mIU/ml) 5,000 - 50,000 **OR**
AFP (ng/ml) 1,000 - 10,000
- *S3 LDH > 10 X N **OR**
hCG (mIU/ml) > 50,000 **OR**
AFP (ng/ml) > 10,000

* Patient should be placed in the category corresponding to the highest level that they satisfy (i.e., use highest relative marker value for categorizing).

N indicates the upper limit of normal for the LDH assay.

5.0 ELIGIBILITY CRITERIA

Each of the criteria in the following section must be met in order for a patient to be considered eligible for registration. Use the spaces provided to confirm a patient's eligibility.

- ___5.1 Patients with prior histologic or serologic confirmation of a testicular or extragonadal germ cell neoplasm not amenable to surgical cure.
- ___5.2 Patient must have refractory germ cell neoplasms defined as one or more of the following:
- ___1) Patients who progress during or within 4 weeks of cisplatin-containing therapy **OR**
 - ___2) Patients who have failed initial poor risk or salvage chemotherapy regimens, including:
 - high-dose therapy (chemotherapy with stem cell support)
 - POMB-ACE tx
 - VeIP
 - VIP

Definition of Progression:

Progression will be defined as:

- I. Appearance of new metastatic lesion on chest x-ray or computed tomography scan or
- II. Increase of known locally advanced or metastatic disease by RECIST criteria or
- III. Rise in tumor markers (Beta-HCG or AFP) by 50% or more relative to the nadir, (in the absence of a physician defined marker leak phenomena in the first 8 weeks of therapy on the protocol, where the markers may rise. If this is suspected then ongoing therapy must be discussed with the study principal investigator).

Patients with rising tumor markers in the face of no new evaluable lesions will require the following to rule out confounding disease not amenable to standard chemotherapeutics or better treated with other modalities including surgery/XRT:

- CT/MRI brain to rule out metastasis not amenable to conventional chemotherapy.
- Re-evaluation of elevated HCG due to:
 - (a) radioimmuno-assay cross reactivity with lutenizing hormone (LH)²⁶
 - (b) in association with marijuana use²⁷
- Ultrasound of testicle(s) to rule out:
 - (a) contralateral primary
 - (b) testicular disease previously missed and labeled as extragonadal primary

When lesions slowly progress in the face of radiographically during chemotherapy treatment despite an appropriate serologic response with declining tumor markers, the following must be ruled out:

- a. Growing Teratoma Syndrome
This is accomplished by continuation of chemotherapy and subsequent resection of all masses associated with radiologic abnormalities.²⁸
- b. Pulmonary nodules due to Bleomycin –
Seen in patients responding serologically and radiologically in other areas. This will be ruled out with surgical resection of selected residual lung masses when tumor markers have normalized.²⁹

Treatment will continue while there is observed serological response during treatment on protocol. When the only evidence of germ cell progression or recurrence prior to entry is the appearance of a new lesion(s) in the absence of tumor marker elevation, a biopsy will be required to confirm the diagnosis.

- ___ 5.3 Patient must have one or more of the following (check all that apply).
- ___ 1) Unidimensionally measurable disease as defined in Section 10.1a)RECIST criteria) which has been assessed within 28 days prior to registration (Note: Given the potential for rapid growth of these lesions in GCT patients, it is highly desirable to have patients commence treatment with 14 days of imaging studies) **AND/OR**
 - ___ 2) Elevated beta-HCG (> 20 mIU/ml) which has been assessed within 7 days prior to registration **AND/OR**
 - ___ 3) AFP > 2 x institutional upper limit of normal which has been assessed within 7 days prior to registration.
- *NOTE:** Soft tissue disease, which has been radiated in the 2 months prior to registration, is not assessable as measurable disease. Serum markers should be repeated immediately prior to commencement of therapy for baseline but not for eligibility. purposes
- ___ 5.4 X-rays, scans or physical examinations for non-measurable disease must have been completed within 28 days of registration. (Note: Given the potential for rapid growth of these lesions in GCT patients, it is highly desirable to have patients commence treatment with 14 days of imaging studies)
- ___ 5.5 Patient must **not** have had treatment with cytotoxic or experimental agents within 14 days prior to registration.
Date last treatment regimen with cytotoxic or experimental agents stopped _____
- ___ 5.6 Patient may have received prior surgery/radiation therapy. At least 2 weeks must have elapsed since completion of previous therapy and patient must have recovered from any adverse effects.
- ___ 5.7 Patients must have an AGC of $\geq 1,500/\text{mm}^3$, a platelet count of $\geq 100,000/\text{mm}^3$.
AGC _____ Platelet count _____
- ___ 5.8 Patients must have a serum creatinine ≤ 2.5 x institutional upper limit of normal (IULN) or a calculated or measured creatinine clearance ≥ 40 ml/min obtained within 14 days prior to

registration. Patient must not be on renal dialysis.

Calculated creatinine clearance =

(a) Jelliffe Formula $M : Clcr = (98 - (0,8 \times (age - 20))) / \text{serum creatinine}^{30}$

$$F : Cl_{cr} = 0.9 \times (98 - (0.8 \times (\text{age} - 20))) / \text{serum creatinine}$$

Creatinine Clearance

- 5.9 Patient must have a serum total bilirubin $< 2.5 \times \text{IULN}$ and SGOT and alkaline phosphatase $< 5 \times \text{IULN}$ obtained within 14 days prior to registration.

Total Bilirubin	7.1	g	Date Obtained
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SGOT _____ Date Obtained _____

Alkaline Phosphatase _____ Date Obtained _____

- ____ 5.10 Patient must have a serum potassium and magnesium within the institutional range of normal. These tests must be obtained within 14 days prior to registration.

K
Mg

- 5.11 Patient must have LDH obtained within 7 days of registration.

- 5.12 Patients must have no evidence of concurrent infection.

Temperature >38.5°C and WBC > 11,000 unless these values can be ascribed to another tumor related phenomena.

Temperature WBC

- 5.13 Patients must have a Zubrod performance status ≤ 2 (see Section 10.4).

- 5.14 Patients must have passed their 16th birthday.

- 5.15 No other prior malignancy is allowed except for the following: adequately treated basal cell or squamous cell skin cancer, adequately treated Stage I or II cancer from which the patient is currently in complete remission, or any other cancer from which the patient has been disease-free for 5 years.

- 5.16 Men of reproductive potential must have agreed to use an effective contraceptive method.

- ☐ 5.17 All patients must be informed of the investigational nature of this study and must sign and give written informed consent in accordance with institutional and federal guidelines.

6.0 STRATIFICATION FACTORS

Patients will not be stratified under this treatment regimen.

7.0 TREATMENT PLAN

- ### 7.1 Dose and schedule:

Patients will receive treatment beginning Week 1 and every two weeks thereafter with following regimen:

Treatment REGIMEN consists of paclitaxel 170mg/m² IV over three hours followed by gemcitabine 800 mg/m² IV at a rate of 10 mg/m²/minute (that is, 80 minutes) and finally oxaliplatin 100 mg/m² IV over 90 minutes .

AGENT	DOSE	ROUTE	DAYS	INTERVAL
1)Paclitaxel	170 mg/m2	IV	d 1	14 days
2)Gemcitabine	800 mg/m2	IV	d 1	14 days
3)Oxaliplatin	100 mg/m2	IV	d 1	14 days

SCHEMA:

A cycle is defined as 14 days of therapy unless there is dose delay when the cycle counts will be based on the number of times the patient commences the above therapy.

Paclitaxel

Gemcitabine

Paclitaxel

Gemcitabine

Oxaliplatin

Paclitaxel premedication consists of dexamethasone 20mg (oral or IV), diphenhydramine 50mg IV, and famotidine 20 mg IV. Anti-emetics at any time are permissible during treatment and should be given as per institutional policy. Replacement of serum electrolytes may be undertaken at any time during treatment.

Prehydration/Posthydration: 100 milliliters D5W (5% dextrose) + 1 gram Calcium Gluconate + 1 gram Magnesium SO₄ to be given IV over 30 minutes pre and post Oxaliplatin

7.2 Delay of treatment

Patients will continue treatment every 2 weeks until disease progression or other reason for discontinuation of protocol treatment.

Day 1 of a new 14 day treatment cycle will begin when absolute neutrophil count is $\geq$ 1,000/mm³ and the platelet count $\geq$ 75,000/mm³. **EXCEPTION:** The physician may treat the patients as scheduled with an absolute neutrophil count of $\geq$ 700/mm³ provided there is monocytosis indicative of marrow recovery and the patient has no manifestations suggesting infection including fever $>38.5^{\circ}\text{C}$.

7.2.1 Concurrent Infection

Patients must have no evidence of concurrent infection.
(temperature $>38.5^{\circ}\text{C}$)

Infection must be proven with appropriate cultures. If fever continues without positive cultures, treatment will continue as planned after appropriate work-up.

Fever can be attributed to multiple other etiologies including tumor fever and related to gemcitabine infusion.

7.3 Criteria for removal from protocol treatment

7.3.1 Patients will discontinue treatment after disease progression at any time (as defined in Section 10.2.4)

7.3.2 Unacceptable toxicity (as defined in Section 8.3).

7.3.3 Delay of treatment of more than 4 weeks due to toxicity or any other reason.

7.3.4 Confirmed CR or confirmed PR lasting > 12 weeks with no further reduction in tumor measurements or tumor markers over at least 8 weeks.

7.3.5 The patient may withdraw from the study at any time for any reason.

7.4 Dose escalation associated with Oxaliplatin

Inherent in this protocol is a one-time opportunity to dose escalate the platinum compound Oxaliplatin in an attempt to maximize synergism between the three chemotherapeutic modalities. This action may also increase response in patients with residual cisplatin sensitivity.

Dose escalation will proceed as follows:

*** Only on Day #14 post cycle #1, that is, Cycle 2, day 1.**

Parameter	Criteria	Modification
AGC and Platelets	> 1500 AND $> 75,000$ AND Neurotoxicity $<$ Grade II AND No evidence of infection AND No evidence of progression of disease	Increase Oxaliplatin by 25% to 125mg/m ² ONE TIME ONLY

*** Dosages will be returned to pre-adjustment values if same criteria are not met on day 14 of post adjustment dosing.**

7.5 Discontinuation of treatment must be documented with the following:

7.5.1 Date of discontinuation.

7.5.2 Reason for discontinuation

7.5.3 Follow-up and future treatment planned.

All responding patients will be followed for a minimum of 3 years from registration to assess duration of remission and date of possible relapse.

8.0 TOXICITIES TO BE MONITORED AND DOSAGE MODIFICATIONS

8.1 Safety

US Regulations require that a sponsor reports Serious Adverse Events (SAEs) occurring with use of its product in a clinical trial if it is unexpected, and felt to be related to use of the drug. During the conduct of Investigator Initiated Trials (IITs), all SAEs that occur will be evaluated by the investigator for reportability to FDA. An Adverse Event should be identified, Serious Adverse Event and Expectedness determined and causality assessed by the investigator using the definitions that follow:

Definitions

An **Adverse Event** is any untoward medical occurrence in a patient administered a pharmaceutical product and which does not necessarily have to have a causal relationship with the treatment.

A **Serious Adverse Event** is any untoward medical occurrence that at any dose:

- Results in death, or
- Is life-threatening, or
- Requires in-patient hospitalization or prolongation of existing hospitalization, or
- Results in persistent or significant disability/incapacity, or
- Is a congenital anomaly/birth defect, or
- Is a medically important event:

Medical and scientific judgment should be exercised in deciding whether expedited reporting is appropriate in other situations, such as important medical events that may not be immediately life-threatening or result in death or hospitalization but may jeopardize the patient or may require intervention to prevent one of the other outcomes listed in the definition above. These should also usually be considered serious.

Examples of such events are intensive treatment in an emergency room or at home for allergic bronchospasm; blood dyscrasias, convulsions that do not result in hospitalization, or development of drug dependency or drug abuse.

Note: The term "life-threatening" in the definition of "Serious Adverse Event" refers to an event in which the patient was at risk of death at the time of the event; it does not refer to an event that hypothetically might have caused death if it were more severe.

An **Unexpected Adverse Event** is not listed in the current Clinical Investigator's Brochure (CIB) / Package Insert or an event that may be mentioned in the CIB / Package Insert, but differs from the event because of greater severity or specificity.

For SAEs occurring in the arm investigating oxaliplatin, the Expectedness of an Adverse Event is determined by identifying the term in the table of Expected Adverse Events by Body System, provided in the Clinical Investigator's Brochure SR96669-EN-E05 or an updated version. If the term is not contained in the list under the relevant body system, the term is to be considered UNEXPECTED.

For comparative drugs, expectedness is determined by using the pertinent reference text: the US Package Insert (if marketed drug) or Investigator Brochure (if Investigational New Drug).

Causality is a determination of whether there is a reasonable possibility that the drug may have caused or contributed to an adverse event. It includes assessing temporal relationships

dechallenge/rechallenge information, association (or lack of association) with underlying diseases, and the presence (or absence) or a lack of one or more likely causes.

Serious Adverse Events Reportable to the FDA

In the context of an Investigator Initiated Trial being conducted under an IND, CFR 312.32 states the FDA needs to be informed of unexpected and related SAEs as soon as possible or within 15 calendar days. In addition, the investigator must notify the FDA of any unexpected, fatal or life-threatening experience associated with the use of the drug as soon as possible, or within 7 calendar days by telephone or facsimile.

When the principal investigator has determined that a Serious Adverse Event requires reporting to the FDA (unexpected and possibly related to study drug), the following actions must be completed:

Actions towards FDA:

- **TELEPHONE the FDA** immediately (day of awareness), in the case of reportable death or life-threatening events.
- **COMPLETE FDA Form 3500.**
- **SEND the completed 3500 form to the FDA** (preferably by fax) within the timelines mentioned above.
- **ATTACH the photocopy of all examinations** that have been carried out as a result of the reportable SAE and the dates on which these examinations were performed. For laboratory results send also the laboratory normal ranges.

Actions towards Sanofi-Synthelabo (oxaliplatin):

- **Concurrently PROVIDE** a copy to Sanofi-Synthelabo at New York Safety Surveillance by fax **866-692-7371** at the latest within 24 hours after the FDA submission (initial and follow-up information)

Actions towards co/sub-investigators (participating on this protocol)/Ethics Committee (IRB):

- For this trial, the principal investigator is also responsible for providing all Serious Adverse Events (on form 3500) submitted to the FDA to the IRB and co/sub-investigators (Dear Investigator Letter) participating on this protocol. This applies to initial and follow-up information.

Actions towards other manufacturers (as relevant)

1. For reportable Serious Adverse Events (completed form 3500) that are reported in patients treated with products other than oxaliplatin, the principal investigator must also transmit the information to the manufacturer of the suspect product within 24 hours.

Follow-up Reports

- The Investigator should take all appropriate measures to ensure the safety of the patients, notably he/she should follow up the outcome of any Serious Adverse Event submitted to the FDA on a 3500 form and complete follow-up forms as necessary. The patient must be followed up until clinical recovery is complete and/or laboratory results have returned to normal. This may mean that follow-up will continue after the patient has completed the trial and that additional investigations may be necessary.
- Any reportable Serious Adverse Events brought to the attention of the Investigator at any time after cessation of the trial and considered by him/her to be reasonably associated with medication administered during the period should also be submitted to the FDA. As for initial reports, a copy is to be provided to Sanofi-Synthelabo New York Safety Surveillance or the appropriate manufacturer of the product within 24 hours after FDA submission (see above)

- The 3500 form providing follow-up information about a reportable SAE must be submitted to the FDA as soon as possible or within 15 calendar days.
- As with the initial submission to the FDA of the 3500 form, the principal investigator is also responsible for providing all follow-ups for reportable Serious Adverse Events (on form 3500) to the IRB and co/sub-investigators (Dear Investigator Letter) participating on this protocol.

IND Annual Report (if applicable)

The principal investigator is responsible for preparing and submitting the IND annual reports to the FDA (CFR312.33) within 60 days of the anniversary date that the IND went into effect. A copy of the report shall be provided to New York Safety Surveillance within 24 hours of submission to the FDA.

DSMB Meeting Minutes (if applicable)/Other Safety Issues

As appropriate, following each Data Safety Monitoring Board (DSMB) providing a summary of adverse events and DSMB recommendations, a courtesy copy of meeting minutes or other oxaliplatin safety issue reports shall be provided to Sanofi-Synthelabo New York Safety Surveillance, within 24 hours (Fax no: 866-692-7371).

- 8.2 This study will utilize the CTC (NCI Common Terminology Criteria) Version 3.0 for toxicity (as summarized in the USC Toxicity Worksheet) and Adverse Event reporting. A copy of the CTC Version 3.0 can be downloaded from the CTEP home page (<http://ctep.cancer.gov>).

8.3 Unacceptable Toxicity

Dose modifications for toxicity will be based on the maximum toxicity grade observed on day 15 assessment (see section 8.5).

No more than two 25% dose reductions will be permitted. A patient who experiences a Grade 3 or higher toxicity after two dose reductions will be removed from the study. Dose re-escalations are not allowed.

Unacceptable toxicity for the purposes of this study will be defined as treatment related death, grade 4 non-hematological toxicity not resolving within 7 days, grade 4 hematological toxicity requiring platelet transfusion or hospital admission for treatment of neutropenic fever over 38.5 degrees Celsius or requirement for more than 2 doses reductions for toxicity.

8.4 G-CSF (Amgen)

G-CSF has been licensed by the Food and Drug Administration for the prevention of chemotherapy induced neutropenia. In this study, G-CSF will not be administered to all patients to prevent neutropenia. However, it may be used for patients who develop neutrophil counts <500/mm³, experienced neutropenic fever, or documented infections while neutropenic.

For patients who experience (1) Grade 3 or 4 neutropenia or (2) develop neutropenic fever between cycles of chemotherapy, G-CSF may be added to all subsequent cycles of chemotherapy. If G-CSF is used, it is recommended that it be used in the following manner:

If a patient develops neutropenia following chemotherapy, all dose modifications outlined in the protocol will be followed according to the original protocol. G-CSF will be added to all subsequent cycles of chemotherapy, unless there is clinical suspicion that the neutropenia was due to an unrelated medical condition and not due to the chemotherapy.

G-CSF will be given at a dose of 5 µg/kg/d subcutaneously beginning 24 hours after completion of chemotherapy and continuing until the AGC exceeds 10,000/µl on two successive determinations. G-CSF should be discontinued at least 24 hours before the next chemotherapy dose. While the

patient is receiving G-CSF, the CBC should be monitored at least once a week (more frequently if clinically indicated). The use of G-CSF and any toxicities related to G-CSF must be documented.

Pegulated G-CSF (Neulasta) may also be used as an alternative to G-CSF provided institutional guidelines are adhered to. Erythropoietin use for anemia is also acceptable.

8.5 Dose Modifications.

If toxicities are Grade 1, they should be managed symptomatically, if possible, and treated without dose reduction.

If toxicities are greater than Grade 1, treatment should be withheld (except for anemia or neuropathy or as specifically indicated below) until recovery to Grade 1 (or baseline, if baseline was > Grade 1).

Dose modifications for the next cycle and additional cycles are based on day 15 complete blood count with differential and toxicity work sheet unless the patient develops cytopenia associated infection or bleeding during the prior cycle (see 8.5.1 below) or requires hospital admission at which time a toxicity assessment should be undertaken and CBC obtained.

- 8.5.1 Hematologic On day 15, if ANC < 700 or 700 - 1,000 without monocytosis, or platelets < 75,000, then therapy will be withheld until ANC $\geq$ 1,000 and platelets $\geq$ 75,000. Patients with ANC < 700 or 700 - 1,000 without monocytosis will resume therapy at paclitaxel and gemcitabine doses reduced by 25% for all subsequent cycles.

Time Point	Parameter	Criteria	Decrease dose
Day 14 in each cycle	AGC and/or Platelets	700 - 1000 without monocytosis 75,000 - 99,999	Paclitaxel by 25%; Gemcitabine by 25%
Day 14 in each cycle	AGC and/or Platelets	<700 <75,000	Paclitaxel by 25%; Gemcitabine by 25%
Any time during cycle	AGC	<500 + fever >38.5°C	Gemcitabine by 25% Paclitaxel by 25%
Any time during cycle	Platelets	<75,000 + bleeding	Gemcitabine by 25%; Paclitaxel by 25%

- 8.5.2 Hepatic: Patients with Grade 4 hepatic toxicity will have treatment held until toxicity recovers to <Grade 2 or below.

- 8.5.3 Stomatitis: If Grade 3/4 stomatitis occurs, hold until recovery to Grade $\leq$ 1.

- 8.5.4 Oxaliplatin Hypersensitivity reactions

Allergic reactions

Following a Grade 1 or 2 acute hypersensitivity reaction that is assessed as related to Eloxatin™ administration, the following premedication should be administered prior to each subsequent dose of Eloxatin™ (patients who have grade 3 or 4 acute hypersensitivity reactions should discontinue Eloxatin™ therapy):

Dexamethasone 20 mg PO or IV, 12 and 6 hours prior to the Eloxatin™ dose;
Dexamethasone 20 mg PO or IV, as well as diphenhydramine 50 mg IV, and one of the following: cimetidine 300 mg IV, ranitidine 50 mg IV, or famotidine 20 mg IV 30-60 minutes prior to Eloxatin™ administration.

- 8.5.5 Paclitaxel Hypersensitivity reactions:

Treatment shall be discontinued for Grade 4 hypersensitivity reactions.
There are no dose reductions for hypersensitivity reactions.

Hypersensitivity Treatment Guidelines**Grade 1**

- Consider decreasing the rate of infusion until recovery from symptoms; closely monitor patient.
- Resume paclitaxel at the planned initial rate after symptoms resolve on that day of treatment.

Grade 2

- Interrupt infusion immediately.
- Give diphenhydramine 50 mg IV with or without dexamethasone 10 mg IV; monitor patient until symptoms resolve.
- Resume infusion after recovery of symptoms; depending on the physician's assessment of the patient, infusion should be resumed at a slower rate, then increased incrementally to the initial planned rate.
- Depending on the intensity of the reaction observed, additional oral or IV premedication with an antihistamine should also be given for the **next cycle** of treatment, and the rate of infusion should be decreased initially and then increased back to the recommended 1-hour.

Grade 3

- Immediately discontinue infusion.
- Give diphenhydramine 50 mg IV with or without dexamethasone 10 mg IV and/or epinephrine as needed; monitor patient until resolution of symptoms.
- Resume infusion after recovery of symptoms on that day of treatment; depending on the physician's assessment of the patient, infusion should be resumed at a slower rate, then increased incrementally to the initial planned rate.
- Depending on the intensity of the reaction observed, additional oral or IV premedication with an antihistamine should also be given for the **next cycle** of treatment. The rate of infusion should be decreased initially and then increased back to the recommended 1-hour if symptoms have resolved.

Grade 4 • NO FURTHER THERAPY

8.5.7 Peripheral neuropathy:

Neuropathic symptoms and signs: patients may be treated if they have grade 2 neuropathy on day 15 but dose delay will be at the physicians discretion and dose reduction should not be undertaken. For grade 3 or greater neuropathy on day 15, all drugs **must** be held until resolution to grade 2, at which time the patient should have a 25% reduction in doses of oxaliplatin and paclitaxel. Dose modification will NOT be undertaken for transient neuropathic symptoms during the cycle that have resolved by day 15.

8.5.8 Dose adjustments for Toxicities Associated with G-CSF

Dose modifications for G-CSF toxicity (bone pain, splenomegaly, abnormalities in uric acid concentrations, LDH and alkaline phosphatase, transient elevations of serum creatinine and aminotransferase activity) should only be initiated if symptomatic control of the toxicity fails (i.e., analgesics such as acetaminophen or acetaminophen with codeine for myalgias or bone pain, etc.).

Toxicity Grade**Dose Adjustment**

Grade 0 - 1

No change

Grade 2

Decrease G-CSF to 3 µg/kg/d

Grade 3 - 4

Discontinue G-CSF

8.5.9 **Pregnancy Guidance**

If a male patient is suspected of having fathered a child while on study drugs, the pregnant female partner must be notified and counseled regarding the risk to the fetus. In addition, the treating physician must follow the course of the pregnancy, including prenatal and neonatal outcome. Infants should be followed for a minimum of 8 weeks.

Upon live-birth delivery, minimum information that should be collected includes date of birth, length of pregnancy, sex of infant, major and minor anomalies identified at birth. Outcomes can be obtained via mailed questionnaires, maternal interviews, medical record abstraction, or a combination of these methods. All serious adverse event reports relating to the pregnancy, including spontaneous abortion, elective abortion and congenital anomalies, should be forwarded to the FDA and a copy sent to Sanofi-Synthelabo (See Safety reporting section).

8.5.10 Dose escalation associated with Oxaliplatin (see also section 7.5).

Inherent in this protocol is a one-time opportunity to dose escalate the platinum compound Oxaliplatin in an attempt to maximize synergism between the three chemotherapeutic modalities. This action may also increase response in patients with residual cisplatin sensitivity.

Dose escalation will proceed as follows:

*** Only on Day #14 post cycle #1, that is, Cycle 2, day 1.**

Parameter	Criteria	Modification
AGC and Platelets	> 1500 AND > 75,000 AND Neurotoxicity < Grade II AND No evidence of infection AND No evidence of progression of disease	Increase Oxaliplatin by 25% to 125mg/m² ONE TIME ONLY

*** Dosages will be returned to pre-adjustment values if same criteria are not met on day 14 of post adjustment dosing.**

9.0 STUDY CALENDAR

A PHASE II STUDY OF GEMCITABINE, OXALIPLATIN, AND PACLITAXEL IN PATIENTS WITH REFRACTORY GERM CELL CARCINOMA

REQUIRED STUDIES	PRE	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk	Wk
	STUDY	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
PHYSICAL																			
History and Physical Exam	X	X		X		X		X		X		X		X		X		X	
Weight and Performance Status	X	X		X		X		X		X		X		X		X		X	
Disease Assessment	X	X				X				X				X				X	
Baseline Abnormality Assessment	X	X																	
LABORATORY																			
CBC/Differential/Platelets		X		X		X		X		X		X		X		X		X	
Serum Creatinine/Creatinine Cl		X		X		X		X		X		X		X		X		X	
Serum Beta-HCG, AFP, LDH		X		X		X		X		X				X				X	
Serum Magnesium & Potassium		X		X		X		X		X		X		X		X		X	
Serum Calcium		X				X				X				X				X	
Serum Bilirubin/SGOT/ Alk Phos		X				X				X				X				X	
Toxicity Notation		X		X		X		X		X		X		X		X		X	
X-RAYS AND SCANS																			
Xrays/Scans as needed for tumor assessment *	X									X								X	
TREATMENT																			
Gemcitabine (Day 1)		X		X		X		X		X		X		X		X		X	
Paclitaxel (Day 1)		X		X		X		X		X		X		X		X		X	
Oxaliplatin (Day 1)		X		X		X		X		X		X		X		X		X	

Treatment is given on Day 1 of all subsequent cycles every 2 weeks until tumor progression, unacceptable toxicity or other reason for discontinuation.

Laboratory values should be obtained within 32 hours of each 14 day treatment cycle. Before treatment is given on Day 1, the results of these tests must be known.

*Tumor assessment will be done after every 8 weeks +/- 7 days by the same method used at baseline.

RESPONSE MUST BE CONFIRMED BY A SECOND DETERMINATION AT LEAST 4 WEEKS AFTER A COMPLETE OR PARTIAL RESPONSE HAS BEEN NOTED.

1. Protocol treatment and parameters will continue at these intervals until progression of disease, confirmed complete response, or one cycle post confirmed partial response lasting > 12 weeks with no further reduction in tumor or markers for at least 8 weeks.

2. Follow up will continue every 2 weeks for three months and will then be adjusted at the discretion of the investigator according to accepted standards for germ cell cancer patients.

10.0 CRITERIA FOR EVALUATION AND ENDPOINT DEFINITIONS

Response

Patients are to be assessed for response at every cycle for disease measurable by physical examination, standard radiographs of evaluable tumor (i.e. chest and pelvic), and serum tumor markers. Disease requiring x-rays or scans must be assessed at least every other cycle for first 8 weeks of treatment regimen. Biomarkers must be assessed at least once every cycle. Tumor markers will be evaluated weekly on day one of treatment with evaluation of results done at next visit (1 week). Evaluation of tumor response will be reserved and only evaluated at after 2 consecutive treatments (greater than or equal to 4 weeks of protocol treatment).

In the event of complete response (CR) of tumor markers, markers will be resent and one additional cycle of therapy will be given and patient will then be re-evaluated. Tumor markers will be required to stay below the upper limits of normal for at least 4 consecutive weeks.

10.1 Measurability of lesions

10.1.1 Measurable disease: Lesions that can be accurately measured in at least one dimension by:

- 1) medical photograph (skin or oral lesion), palpation, plain x-ray, CT, MRI or other conventional technique with longest diameter 2 cm or greater in the axial plane (bone lesions not included), or
- 2) spiral CT with longest diameter 1 cm or greater.

Ultrasound is suitable only for superficial disease (superficial palpable nodes, subcutaneous lesions, thyroid nodules). Conventional CT and MRI should be performed with cuts of 10 mm or less in slice thickness contiguously. Spiral CT should be performed using a 5 mm contiguous reconstruction algorithm.

10.1.2 Non-measurable disease: All other lesions including lesions too small to be considered measurable, pleural or pericardial effusions, ascites, bone disease, inflammatory breast disease, leptomeningeal disease, lymphangitis, pneumonitis, abdominal masses not confirmed and followed by imaging techniques, cystic lesions or disease documented by indirect evidence only (e.g., by lab values).

10.1.3 Tumor Markers: Serum AFP ($> 2 \times \text{IULN}$) and Beta-HCG ($> 20 \text{ mIU/ml}$) levels are considered measurable disease in germ cell neoplasms.

10.2 Objective status at each evaluation:

Guidelines set forth by RECIST will be used to evaluate all soft tissue lesions.

Objective Status is to be recorded at each evaluation. All measurable lesions up to a maximum of 5 lesions per organ and 10 lesions in total, representative of all involved organs, should be identified as target lesions and recorded at baseline. All measurable lesions not identified as target lesions are non-target lesions and are included as non-measurable disease. Measurements must be provided for target measurable lesions, while presence or absence must be noted for non-target measurable and non-measurable disease.

10.2.1 Complete Response (CR): Complete disappearance of all measurable and non-measurable disease. No new lesions. Complete disappearance of all clinical, radiographic and biochemical (LDH, AFP and Beta-HCG less than institutional upper limit of normal) evidence of disease for a minimum of 4 weeks. No disease related symptoms. Normalization of markers and other abnormal lab values. All disease must be assessed using the same technique as baseline.

10.2.2 Partial Response (PR): In patients with at least one measurable lesion greater than or equal to 30% decrease under baseline of the sum of longest diameters of all target measurable lesions. No unequivocal progression of nonmeasurable disease. No new lesions and no evidence of progression based on AFP and Beta-HCG. All target measurable lesions must be assessed using the same techniques as baseline. In patients who have elevated AFP ($> 2 \times \text{IULN}$) or Beta-HCG ($> 20 \text{ ml U/ml}$)

as the only measurable disease, a 75% reduction from baseline or a reduction to < IULN in each elevated marker must occur in order for there to be a partial response.

10.2.3 Stable: Does not qualify for CR, PR, Progression or Symptomatic Deterioration. All target measurable lesions must be assessed using the same techniques as baseline.

10.2.4 Progression: One or more of the following must occur: 20% increase in the sum of longest diameters of target measurable lesions over smallest sum observed (over baseline if no decrease during therapy) using the same techniques as baseline. Unequivocal progression of non-measurable disease in the opinion of the treating physician (an explanation must be provided). Appearance of any new lesion/site. Death due to disease without prior documentation of progression and without symptomatic deterioration (see Section 10.2.5). Biomarker progression is defined as at least two consecutive rises in tumor marker to be documented over a reference value (measure 1). The third measurement (2nd beyond the reference level) must be greater than the second measure. If the third measure is not greater than the second, then a fourth measurement is required to be taken and be greater than the second measure. The time period encompassing all of these measurements must be at least two weeks. Biomarker progression as defined may occur with either tumor marker (AFP or Beta-HCG) alone.

10.2.5 Symptomatic deterioration: Global deterioration of health status requiring discontinuation of treatment without objective evidence of progression. Efforts should be made to obtain objective evidence of progression after discontinuation.

10.2.6 Assessment inadequate, objective status unknown: Progression or symptomatic deterioration has not been documented, and one or more target measurable lesions have not been assessed or inconsistent assessment methods were used.

10.2.7 Objective status notes:

1. Non-measurable and non-target measurable disease do not affect objective status except in determination of CR (must be absent--a patient who otherwise has a CR, but who has non-measurable or non-target measurable disease present or not assessed, will be classified as having a PR), and in determination of progression (if new sites of disease develop or if unequivocal progression occurs in the opinion of the treating physician).
2. An objective status of PR or stable cannot follow one of CR. Stable can follow PR only in the rare case that tumor increases too little to qualify as progression, but enough that a previously documented 30% decrease no longer holds.
3. In cases for which initial flare reaction is possible (hypercalcemia, increased bone pain, erythema of skin lesions), objective status is not progression unless either symptoms persist beyond 4 weeks or there is additional evidence of progression.
4. Lesions that appear to increase in size due to presence of necrotic tissue will not be considered to have progressed.
5. For bone disease documented on bone scan only, increased uptake does not constitute unequivocal progression.
6. Appearance or worsening of pleural effusions does not constitute unequivocal progression unless cytologically proven of neoplastic origin.
7. If CR determination depends on a lesion for which the status is unclear by the required tests, it is recommended the residual lesion be investigated with biopsy or fine needle aspirate.

10.3 Best Response: This is calculated from the sequence of objective statuses.

- a. CR: Two or more objective statuses of CR a minimum of four weeks apart documented before progression or symptomatic deterioration.

- b. PR: Two or more objective statuses of PR or better a minimum of four weeks apart documented before progression or symptomatic deterioration, but not qualifying as CR.
- c. Unconfirmed CR: One objective status of CR documented before progression or symptomatic deterioration but not qualifying as CR or PR.
- d. Unconfirmed PR: One objective status of PR documented before progression or symptomatic deterioration but not qualifying as CR, PR or unconfirmed CR.
- e. Increasing disease: Objective status of progression or symptomatic deterioration within 8 weeks of registration, not qualifying as anything else above.
- f. Stable/no response: At least one objective status of stable/no response documented at least 8 weeks after registration and before progression or symptomatic deterioration, but not qualifying as anything else above
- g. Inadequate assessment, response unknown: Progression or symptomatic deterioration greater than 12 weeks after registration and no other response category applies.

OVERALL BEST RESPONSE		
	MEASURABLE DISEASE	Tumor Markers
Complete Response	Complete Response	Complete Response
	Complete Response	Absent
	Absent	Complete Response
Partial Response	Partial Response	Complete Response
		Absent
		Stable Disease
	Partial Response	Partial Response
	Complete Response	Partial Response
	Absent	
	Stable Disease	
Stable Disease	Stable Disease	Stable Disease
	Stable Disease	Absent
	Absent	Stable Disease

10.4 Performance Status: Patients will be graded according to the Zubrod performance status scale.

POINT DESCRIPTION

- 0 Fully active, able to carry on all pre-disease performance without restriction.
- 1 Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light housework, office work.
- 2 Ambulatory and capable of self-care but unable to carry out any work activities; up and about more than 50% of waking hours.
- 3 Capable of limited self-care, confined to bed or chair more than 50% of waking hours.
- 4 Completely disabled; cannot carry on any self-care; totally confined to bed or chair.

10.5 Progression –free survival:

From date of first treatment to date of first observation of progressive disease (as defined in 10.2.4), symptomatic deterioration (as defined in 10.2.5) or death due to any cause.

10.6 Time to Death: From date of registration to date of death due to any cause.

11.0 STATISTICAL CONSIDERATIONS

Based on University of Southern California clinical practice, we anticipate an accrual rate of 5 to 7 patients per year.

11.1 Study Design

This is a phase II single-center trial of gemcitabine, oxaliplatin, and paclitaxel in patients with refractory germ cell carcinoma. The outcome variables for this study are tumor response, event-free survival, and safety of treatment. The primary objective of this study is to evaluate the overall response rate (complete and partial response rate) in patients with confirmed refractory germ cell cancer. A two-stage design will be used for patient accrual. Secondary responses will be time to progression and overall survival.

Initially we will enroll 10 patients. If 1 or more patients respond, then we will enroll an additional 20 patients (for a total of 30).

11.2 Sample Size Justification

Initially 10 patients will be accrued. If 2 or fewer of these 10 patients experience a major response (CR or PR as defined in Section 10.0), then we will conclude that the true response rate for this regimen is less than 50%. If 3 or more patients respond, then we will enroll an additional 20 patients (total 30). With 10 patients in the initial stage there is at least 94.5% chance that at least one patient will experience a response, if the true response rate is 50% or greater, and with 30 evaluable patients, we will construct an exact, one-sided 95% confidence interval for the binomial probability (i.e. the true probability of response for this

regimen for these patients). If the value of 50% is not contained in this confidence interval, then we will conclude that this regimen does not warrant further study.

The total sample size of 30 was selected because this can realistically be completed in 4 to 5 years and we will obtain useful preliminary information on feasibility of this regimen.

11.3 Statistical Analysis

All variables, including demographic variables (age, gender, race) will be tabulated. For continuous variables, results will be expressed as mean values $\pm$ standard variables. Categorical variables will be summarized using proportions

11.4 Summary of Results

The outcome status (in terms of toxicity, response, reason off study, progression-free survival, and survival) of all eligible patients who are registered, will be reported. Patients who complete 6 weeks of therapy or who experience any Grade 2 or greater toxicity will be included in the analysis of toxicity. Patients with measurable tumor who complete 8 weeks of therapy, or who terminate treatment for reasons of toxicity or progression, will be included in analysis of tumor response, and in any decision to terminate the study early. All eligible patients who begin treatment will be included in the analysis of survival and time to progression.

All response and their durations will be reported. The response rate and a one-sided 95% confidence interval for the response rate will be calculated. Time to progression and overall survival will be estimated using the product-limit method of Kaplan and Meier. Toxicity information recorded will include the type, severity, time of onset, time of resolution, and the probable association with the study regimen. Tables will be constructed to summarize the observed incidence by severity and type of toxicity.

11.5 Data and Safety Monitoring

There is no formal data and safety monitoring committee for Phase II studies. Toxicity and accrual monitoring are done routinely by the study coordinator. Response monitoring is done by the study statistician and study coordinator. This study will utilize the CTC (NCI Common Toxicity Criteria) Version 3.0 for toxicity and Adverse Event reporting. The patient will be assessed at every clinic visit for toxicity. Additionally all radiologic scans will be reviewed by the principal investigator for discrepancies in assessment of disease and measurements. Tumor marker elevations will be confirmed by secondary labs sent at subsequent physician visit.

12.0 Patient registration

12.1. Registration Guidelines

Patients will be registered on this protocol by calling the USC/Norris Clinical Investigations Support Office at (323) 865-0451 and submitting the study name and the USC/Norris or LAC+USC patient identification number. Registration will occur when the USC/Norris data managers receive a complete registration packet consisting of the Informed consent form and the Registration/Eligibility worksheet.

12.2 Registration Forms

One copy each of the Informed Consent form, Registration/Eligibility worksheet, and flow chart should be submitted at time of registration.

13.0 Ethical and Regulatory Considerations

13.1 General:

All institutional, NCI, and Federal regulations concerning the Informed Consent form will be fulfilled.

13.2 Institutional Review Board Approval

It is required that a valid Institutional Review Board approves the conduct of this clinical study, together with the Investigator's informed consent document, prior to study initiation.

This study is IND exempt, however, the minimum standards of conduct and requirements for informed consent defined in the FDA regulations should be adhered to.

This protocol must be reviewed and approved by a valid Institutional Review Board (IRB) prior to initiation of the study. Until written approval by the IRB has been received by the Investigator, no subject may undergo any procedure solely for the purpose of determining eligibility for this study. Protocol amendments must also be reviewed and approved by the IRB.

13.3 Informed Consent

The Investigator will obtain informed consent from each subject enrolled in the study, in accordance with the U.S. Food and Drug Administration (FDA) regulations 21 CFR Parts 50.20 - 50.27 and the laws and regulations of the country in which the investigation is being conducted. The IRB must approve the informed consent document to be used by the Investigator. It is the responsibility of the Investigator to ensure that informed consent be obtained from the subject or his/her guardian or legal representative before any activity or treatment is undertaken which is not part of routine care. This includes, but is not limited to, the performance of diagnostic or therapeutic procedures and the administration of the first dose of study medication.

13.4 Clinical Laboratory Tests and Normal Laboratory Values (NLV)

Clinical laboratory tests should be performed by the same laboratory throughout the study. The normal ranges of the laboratory that will perform the tests required by the protocol must be transmitted to the study coordinators. If the investigator feels the patient requires additional testing at an outside laboratory, the normal ranges of each laboratory that will perform the tests required by the protocol must be transmitted to study coordinators.

13.5 Case Report Forms (CRFs)

Subject source documents are the physician's subject records maintained at the study site. In most cases, the source documents will be the hospital's or the physician's chart. In cases where the source documents are the hospital's or the physician's chart, the information collected on the CRFs must match those charts.

In some cases, a portion of the source documents for a given subject may be the CRFs. It is the Investigator's responsibility to ensure completion and to review and approve all CRFs. If an Investigator Approval Form is used, it must be signed by the Investigator. Individual CRFs may be signed by the Investigator or a Subinvestigator. These signatures serve to attest that the information contained on the CRFs is true. At all times, the Investigator has final responsibility for the accuracy and authenticity of all clinical and laboratory data entered on the CRFs. The CRFs will be printed on NCR ("no carbon required") paper to permit multiple copies. The bottom copy is to be retained at the site for the Investigator's study file. All questions should be answered using a black ink ballpoint pen. If certain data are not available, not done, or not applicable: "NAV," "ND," or "NAP," respectively, will be entered in the appropriate space.

Twenty-four hour clock should be used for all time entries. (NOTE: 24:00 hours will be entered as 23:59.) Changes and/or additions to data entered on original CRFs must be made in the following manner: The original entry will be lined out with a single line drawn through the error (not erased or "whited out") so as to leave it still legible. The correction will be entered using a black ink ballpoint pen, initialed, and dated by the person making the correction. The Investigator or delegate (e.g., Sub investigator or study coordinator) may enter corrections on original CRFs. The monitoring team may make changes to the copies of CRFs based on information supplied by the Investigator and documented in the study file.

13.6 Disposition of Clinical Supplies

Commercial supplies of drugs will be used and standard records maintained by Norris and USC-LAC pharmacies accessible for review, if required.

13.7 Maintenance of Records

The Investigator will retain a copy of all study documents, including reports to the IRB in accordance with the FDA or local regulations, whichever are the more stringent.

If the Investigator relocates, retires, or for any reason withdraws from the study, the study records may be transferred to an acceptable designee, such as another Investigator.

13.8 Changes in Protocol

Changes to the protocol that affect the decision of the IRB (e.g., more extensive procedures, increased risk to subjects, changes in the subject population, additional safety information, etc.) must be documented in the form of an amendment. If the amendment is minor or reduces the risk to the subject, the chairperson of the IRB alone may approve it. The only circumstances in which the amendment may be initiated without IRB approval is where the change is necessary to eliminate apparent immediate hazards to the subjects. In that event, the Investigator must notify the IRB in writing within five (5) working days after the implementation.

14.0 MINORITIES AND GENDER STATEMENT

Recruitment is open to all minorities and both genders. Although distributions may vary by disease type, our recruitment procedures have been developed to enroll patients who are representative of the respective target population. The ethnic and gender distribution of cancer patients in the counties of the participating institution

Inclusion of Minorities:

This study will include patients of any world group, but is not designed to measure differences of intervention effects between population groups. As 30-40% of patient cases with refractory germ cell cancers are Latino, a significant population of minority cases will be included. It should be noted that Asian and African American males have a lower incidence of testicular cancer.

15.0 ETHICAL AND REGULATORY CONSIDERATIONS

All patients will have signed an informed consent for participation in research activities in accord with all institutional, NCI and Federal regulations, and will have been given a copy of the Experimental Subject's Bill of Rights.

16.0 PATHOLOGY REVIEW

All patients will have malignancy confirmed by review of their biopsy specimens by the Division of Pathology of the University of Southern California/ LA County/ Norris Comprehensive Cancer Center.

17.0 REFERENCES

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